
Robust Low-Light Image Enhancement via Generative Priors: A CSCE753 Project Final Report

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1 Introduction

Images captured under poor lighting conditions often suffer from severely reduced visibility and diminished detail, significantly impairing their usefulness in important applications such as autonomous driving, remote sensing, surveillance, and medical imaging. Enhancing these images to recover clear and natural details is critical, yet challenging, particularly when paired examples (i.e., corresponding well-lit and poorly-lit images) are unavailable or impractical to obtain.

Traditional methods for low-light image enhancement frequently rely on supervised training approaches that require paired datasets or handcrafted techniques based on Retinex theory. However, these strategies often fall short in real-world conditions where capturing ideal reference images is difficult or impossible. As a result, there is a pressing need for more flexible methods that can effectively restore image clarity without depending on explicit ground-truth examples.

Motivated by this challenge, we explore a recent state-of-the-art approach known as *latent-Retinex diffusion*, initially proposed in the LightenDiffusion paper[1]. Our primary goal is to replicate and improve upon the promising results reported in this approach. Additionally, this project aims to build practical expertise in developing, training, and systematically monitoring complex neural network architectures. To facilitate stable training and effective learning, we introduce a carefully structured curriculum-learning approach, where tasks are gradually presented to the model from simpler to increasingly complex.

Contributions

- A structured curriculum-learning strategy designed to stabilize training by sequentially introducing different tasks and objectives.
- Integration of GradNorm, a dynamic loss-balancing technique, combined with gradient monitoring to maintain balanced and stable multi-task training.
- Empirical evaluation demonstrating enhanced image-quality metrics and training stability compared to baseline approaches without curriculum learning.

Report Outline. The remainder of this report is organized as follows: Section 2 summarizes relevant literature; Section 3 details our proposed method and curriculum strategy; Section 4 presents experimental results and detailed analyses; finally, Section 5 concludes the report and suggests potential directions for future research.

Code and Resources. All code and documentation are publicly available at: <https://github.com/omarkhater/lle-generative-priors>.

2 Literature Review

Classical and Retinex-based Approaches. Early methods for low-light image enhancement often leveraged the Retinex theory [2], which assumes that an image can be decomposed into reflectance

(surface information) and illumination (lighting conditions). Methods such as LIME [3] and KinD [4] have applied Retinex-inspired decompositions to explicitly separate illumination from reflectance components, then adjusted the illumination map independently to enhance brightness. Despite their intuitive design and relative computational efficiency, these methods typically rely on heuristic-based assumptions and handcrafted constraints, leading to inconsistent performance across diverse real-world conditions.

Learning-based Methods. To address these limitations, recent approaches have employed deep learning frameworks. Supervised methods, such as RetinexNet [5] and KinD++ [6], rely on paired low-light and well-exposed image datasets to directly learn mappings from degraded to enhanced images. While effective, the requirement for extensive paired training data significantly limits their practicality, as collecting ground-truth reference images in real scenarios is challenging.

Consequently, unsupervised learning-based approaches emerged as attractive alternatives. Enlighten-GAN [7], a generative adversarial network (GAN)-based method, learns image enhancement from unpaired training data, bypassing the need for reference images. Similarly, Zero-DCE [8] proposed a self-supervised approach that learns enhancement curves directly from the input image through a differentiable curve-estimation network. These methods showed improved adaptability to real-world low-light scenarios but sometimes produced artifacts or unnatural enhancements due to the complexity of adversarial training or curve-fitting limitations.

Diffusion-based Image Enhancement. More recently, diffusion probabilistic models (DPMs) [9, 10] have revolutionized image generation tasks. These methods iteratively denoise randomly perturbed data, progressively recovering high-quality outputs. LightenDiffusion [1] adapted DPMs for low-light enhancement, combining Retinex decomposition principles with latent-space diffusion models to improve realism and image quality without the need for paired datasets. Their approach demonstrated substantial qualitative and quantitative improvements over prior unsupervised methods, providing the foundation and motivation for this project.

Curriculum Learning and GradNorm. Curriculum learning, introduced by Bengio et al. [11], advocates training machine learning models on progressively complex tasks, starting with simpler examples and gradually introducing harder ones. This method is known to stabilize training and accelerate convergence in challenging optimization landscapes. Moreover, GradNorm [12] is a gradient-based approach designed to balance competing multi-task objectives dynamically. It achieves this by adaptively tuning loss weights based on gradient norms, preventing any one task from dominating the learning process at the expense of others.

Although curriculum learning and GradNorm have been successfully applied in various computer vision contexts, including semantic segmentation and multi-task classification, their application to low-light image enhancement—particularly within diffusion-based frameworks—is still largely unexplored. Our work addresses this gap by systematically integrating and evaluating curriculum learning and GradNorm strategies within the latent-Retinex diffusion pipeline.

3 Proposed Method

Departures from the Original Design. While the original LightenDiffusion architecture uses a joint optimization of decomposition and reconstruction losses from the start, our approach introduces a structured *curriculum learning strategy* to sequentially activate objectives, stabilizing training and improving convergence. We also integrate *GradNorm* [12] to adaptively balance task weights, replacing manually tuned loss coefficients. Finally, although the original paper introduces composite losses in prose, our implementation formalizes these into a modular, extensible loss registry—facilitating analysis, ablation, and visualization. These changes, originally motivated by training difficulties, evolved into core design decisions.

3.1 Overview of the Two-Stage Framework

We follow a hierarchical structure comprising two distinct stages, illustrated in Figure 1:

1. **Stage 1 (Decomposition and Reconstruction):** This stage trains a Content-Transfer Decomposition Network (CTDN) along with an image reconstruction network. The goal is

to disentangle content and illumination components of low-light images, facilitating better conditions for enhancement.

2. **Stage 2 (Latent-Retinex Diffusion):** Leveraging the latent representations learned in Stage 1, we fine-tune a latent diffusion model to progressively denoise and enhance images, producing realistic and natural illumination.

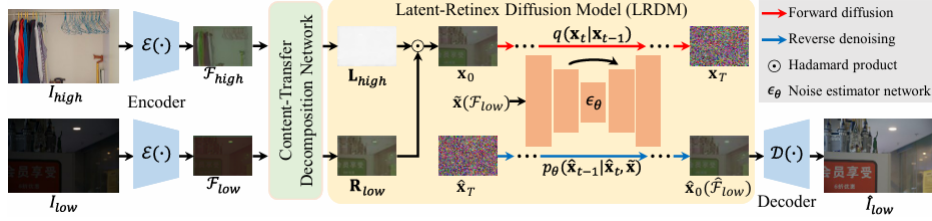


Figure 1: Overview of the proposed two-stage low-light image enhancement framework. Stage 1 learns content and illumination decomposition, while Stage 2 employs latent-Retinex diffusion for image enhancement. [1]

3.2 Curriculum Learning for Stable Training

During preliminary experiments, simultaneous training of the decomposition and reconstruction objectives in Stage 1 frequently led to instability and suboptimal convergence. To mitigate these challenges, we propose a curriculum learning strategy inspired by Bengio et al. [11], systematically introducing tasks from simpler to more complex phases.

- **Phase 1: Content-only training.** Initially, the network is trained solely to reconstruct image content without explicitly modeling the illumination decomposition. This simplifies early optimization, stabilizing network parameters.
- **Phase 2: CTDN-only training.** The network parameters from Phase 1 serve as initialization. We then train the decomposition task exclusively, where the network learns illumination and reflectance disentanglement without requiring content reconstruction.
- **Phase 3: Joint optimization.** Finally, content and decomposition tasks are optimized jointly. By incrementally introducing complexity, we found this approach significantly reduces training instability.

Each phase transition is controlled by predefined epoch milestones, allowing the network to gradually adapt to increasingly complex objectives.

Pseudocode: Curriculum-Aware Stage 1 Trainer

To clarify how curriculum learning is integrated into Stage 1 training, we summarize the key logic in pseudocode form below.

3.3 Dynamic Loss Balancing via GradNorm

Multi-task learning introduces the challenge of balancing multiple loss objectives. To dynamically balance these competing objectives, we incorporate GradNorm [12], an adaptive gradient-normalization method that automatically scales loss weights based on gradient norms. Specifically, given task losses $\mathcal{L}_{content}$ and $\mathcal{L}_{CTDN}$, the overall objective is computed as:

$$\mathcal{L}_{total} = w_{content}\mathcal{L}_{content} + w_{CTDN}\mathcal{L}_{CTDN} \quad (1)$$

GradNorm dynamically adjusts the weights $w_{content}$ and w_{CTDN} based on gradient magnitudes. Formally, the update rule for loss weight w_i for task i is:

Algorithm 1 Curriculum and GradNorm Training Loop

```
1: for epoch = 1 to num_epochs do
2:   if content loss plateaued and not ramp started then
3:     Start CTDN ramp at current epoch
4:   end if
5:   Apply curriculum-based loss weights for this epoch
6:   for each batch in train_loader do
7:      $outputs \leftarrow \text{model}(low\_imgs)$ 
8:      $loss_{total}, raw, weighted \leftarrow \text{calculate\_loss}(outputs)$ 
9:     Backpropagate  $loss_{total}$ 
10:    if step mod gradnorm_interval = 0 then
11:      Backpropagate GradNorm loss
12:    end if
13:    Perform optimizer step
14:    Update log task weights with weight optimizer
15:    Clamp log weights for numerical stability
16:  end for
17: end for
```

$$w_i^{(t+1)} = w_i^{(t)} - \alpha \left(\frac{\|\nabla_{\theta} \mathcal{L}_i\|_2 - \bar{G}}{\bar{G}} \right), \quad \text{where} \quad \bar{G} = \frac{1}{T} \sum_{j=1}^T \|\nabla_{\theta} \mathcal{L}_j\|_2 \quad (2)$$

Here, α is the GradNorm learning rate, $\nabla_{\theta} \mathcal{L}_i$ denotes the gradient of task i 's loss with respect to network parameters θ , and T is the number of tasks (here, $T = 2$).

Algorithm 2 GradNorm Loss Balancing

```
1: Initialize raw loss references  $L_0[i]$  for each task  $i$ 
2: for each training step do
3:   for each task  $i$  do
4:     Compute gradient norm  $g[i] \leftarrow \|\nabla_{\theta}(w_i \cdot \mathcal{L}_i)\|$ 
5:     Compute relative loss rate  $r[i] \leftarrow \left( \frac{\mathcal{L}_i}{L_0[i]} \right)^{\alpha}$ 
6:   end for
7:    $C \leftarrow \frac{1}{T} \sum_j g[j]$ 
8:   for each task  $i$  do
9:      $target[i] \leftarrow C \cdot r[i]$ 
10:  end for
11:   $\mathcal{L}_{gradnorm} \leftarrow \sum_i (g[i] - target[i])^2$ 
12:  Backpropagate  $\mathcal{L}_{gradnorm}$  and update  $w_i$ 
13: end for
```

3.4 Gradient Monitoring

To ensure balanced optimization and quickly identify potential instability during training, we perform continuous gradient norm monitoring. By logging gradient norms for each task at every training iteration, we visualize their evolution throughout training. This practice helps us verify that no single task dominates the optimization process, maintaining effective multi-task learning. Detailed gradient norm results and analysis are presented in Section 4.

4 Experimental Results

This section presents a narrative-based diagnostic investigation into the training challenges encountered while reproducing the LightenDiffusion framework under constrained data and compute settings. While quantitative comparisons are reported where available, the primary goal of this section is to

document the problem-solving trajectory that led to our final curriculum-enhanced pipeline and its limitations.

4.1 Evaluation Metrics

To evaluate the effectiveness of our enhancement models, we report both full-reference and no-reference metrics, as used in prior LLIE benchmarks. Each metric captures a different aspect of perceptual or structural quality:

- **PSNR (Peak Signal-to-Noise Ratio)**: Measures the pixel-wise fidelity between the enhanced image $\hat{I}$ and the reference ground-truth I . It is defined as:

$$\text{PSNR}(\hat{I}, I) = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{\text{MSE}(\hat{I}, I)} \right), \quad (3)$$

where $MAX_I = 1.0$ (for normalized images) and MSE is the mean squared error between $\hat{I}$ and I .

- **SSIM (Structural Similarity Index)**: Quantifies similarity in luminance, contrast, and structure between two images. The SSIM between patches x and y is:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}, \quad (4)$$

where μ , σ , and σ_{xy} are local means, variances, and cross-covariance respectively, and c_1 , c_2 are stability constants.

- **LPIPS (Learned Perceptual Image Patch Similarity)**: Measures perceptual similarity using features extracted from pretrained deep networks (e.g., VGG). It computes:

$$\text{LPIPS}(\hat{I}, I) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|w_l \odot (f_l^h(\hat{I}) - f_l^h(I))\|_2^2, \quad (5)$$

where f_l denotes deep features from layer l and w_l are learned weights.

- **NIQE (Naturalness Image Quality Evaluator)**: A no-reference metric that fits a multivariate Gaussian model to natural image statistics and measures deviation from that distribution:

$$\text{NIQE}(\hat{I}) = \|\mu_{\hat{I}} - \mu_n\|_{\Sigma^{-1}}^2, \quad (6)$$

where $\mu_{\hat{I}}$ is the mean of features from the test image, μ_n and Σ are the mean and covariance of features from pristine natural images.

These metrics were selected based on their complementary strengths. PSNR and SSIM evaluate fidelity to a reference, LPIPS captures perceptual realism, and NIQE penalizes unnatural textures. All metrics were logged via MLflow during training and computed on test images for final reporting.

4.2 Experimental Setup

Datasets. We conduct experiments on the SICE dataset, which includes paired low-light and high-quality images, allowing us to benchmark restoration quality using full-reference metrics. The dataset was split into 219 training, 46 validation, and 30 test samples.

Metrics. Following LLIE benchmarks, we report:

- Full-reference metrics: PSNR, SSIM, LPIPS
- No-reference metric: NIQE

Higher PSNR/SSIM and lower LPIPS/NIQE indicate better perceptual and structural quality.

Implementation. All models were implemented in PyTorch 2.6, tracked using MLflow, and trained on a single NVIDIA RTX3090 GPU. Loss weights were optimized using GradNorm and curriculum scheduling.

4.3 From Heuristics to Learning: Baseline and Paper Checkpoint

We began with a classical image enhancement baseline using non-learned pixel-level transformations (Figure 2). This model applies sequential gamma correction ($\gamma = 0.7$), brightness/contrast/saturation/sharpness boosts, and Contrast Limited Adaptive Histogram Equalization (CLAHE) histogram equalization in LAB space. While fast and interpretable, its performance was inconsistent and structurally limited.

To establish a performance reference, we evaluated the released checkpoint of LightenDiffusion, trained on over 180K images. When run on our SICE-based pipeline, the model achieved perceptually reasonable LPIPS and PI scores, but modest PSNR/SSIM (Table 1).

Table 1: Comparison of baseline, paper model, and best local result on the SICE dataset. All values reflect average over test images. Despite training improvements, the Stage 2 model trained on our best local Stage 1 outputs struggled due to insufficient representation quality, reinforcing Stage 1 as the bottleneck.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	NIQE $\downarrow$	PI $\downarrow$
Non-learned baseline (CLAHE + contrast)	13.42	0.587	0.411	7.02	4.12
LightenDiffusion (official checkpoint)	16.67	0.658	0.289	5.00	2.64
Our best Stage 2	10.37	0.33	0.86	6.22	3.54

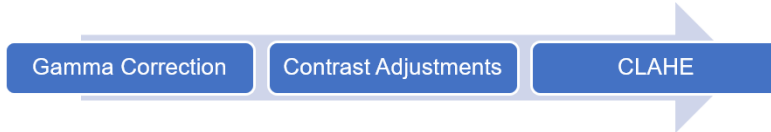


Figure 2: Baseline heuristic pipeline: gamma correction, color boosts, and CLAHE equalization.

4.4 Reproduction Challenges Under Resource Constraints

Attempting to train the original model from scratch on SICE (219 images) led to persistent instability and degraded metrics. Unlike the original setting with large-scale data and extensive compute, we encountered early collapse and overfitting when training with all losses active from the beginning.

4.5 Diagnosing Stage 1: Architecture and Loss Sensitivity

We initially suspected poor initialization or architectural mismatch, and conducted several ablation sweeps:

Architecture Sweep. We evaluated six variations of encoder/decoder widths and attention head configurations. As shown in Table 2, a 64-channel encoder with standard attention (baseline) performed best in terms of PSNR and SSIM. Smaller networks underperformed significantly, while wider or deeper configurations offered negligible benefits and incurred higher cost..

Loss Weight Instability. We ran a 20-run grid search varying λ_g (0.05–10.0) and illumination loss weights (0.1–0.2). As summarized in Table 3, all runs yielded identical PSNR (8.69) and SSIM (0.25), highlighting that manual tuning alone fails to overcome the Stage 1 learning bottleneck. While perceptual metrics such as NIQE and PI varied across configurations, the changes were inconsistent and did not reflect meaningful fidelity improvements. These findings motivated our shift toward curriculum-based progression and dynamic task balancing using GradNorm.

4.6 Diagnosing Stage 2: Sweeps on Loss and Architecture

Before integrating curriculum-trained Stage 1 models, we performed diagnostic sweeps to assess Stage 2 sensitivity to architecture and loss parameters.

Table 2: Stage 1 architecture ablation (fixed weights, no curriculum), sorted by PSNR. Best values in **bold**.

Description	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	NIQE $\downarrow$
Baseline (64 ch)	19.71	0.5995	0.6373	6.80
Mixed attention	19.66	0.5939	0.6585	6.52
More attention	19.43	0.5994	0.6483	6.40
Less attention	19.39	0.5828	0.6687	6.39
Wider network	19.29	0.6046	0.6396	6.07
Smaller network	17.87	0.4999	0.7429	18.77

Table 3: Best and worst loss weight configurations by NIQE and PI (fixed-weight runs). PSNR and SSIM were flat across all runs.

Run ID	λ_g	Illum. Weight	NIQE $\downarrow$	PI $\downarrow$	Notes
sweep_16	0.25	0.2	32.57	32.57	Best perceptual
sweep_18	0.5	0.2	64.33	32.60	Runner-up
sweep_15	0.1	0.2	64.26	32.56	—
sweep_11	0.05	0.1	127.30	64.08	Poorest PI
sweep_14	10	0.1	133.38	67.10	Worst NIQE

Architecture Parameter Sensitivity. Using the top Stage 1 decomposer models (ranked by validation NIQE), we varied DiffusionUNet settings such as the number of residual blocks, channel multipliers, dropout rate, and diffusion timesteps. The best-performing setup used 2 residual blocks, $\text{ch_mult} = [1, 2, 3, 4]$, and 1000 diffusion steps. Results are summarized in Table 4.

Table 4: Stage 2 architecture sweep. PI and NIQE across different dropout and diffusion step settings.

Dropout	Timesteps	PSNR $\uparrow$	SSIM $\uparrow$	NIQE $\downarrow$	PI $\downarrow$
0.0	1000	10.10	0.309	6.41	3.63
0.2	1000	10.23	0.317	6.22	3.54
0.1	1000	10.13	0.331	6.19	3.52
0.1	500	10.17	0.309	6.64	3.68

Loss Weight Sweep. We also varied the illumination correction factor γ and the self-consistency loss weight λ_{scc} to balance content fidelity and latent consistency. Table 5 reports results. The best configuration was $\gamma = 0.1$, $\lambda_{scc} = 0.001$, which achieved strong perceptual metrics with consistent convergence.

These insights highlights the importance of stage 1 model quality and how it influence the results from stage 2 training.

4.7 Debugging with Loss and Gradient Monitoring

To better understand the optimization dynamics in Stage 1, we systematically logged both loss values and per-branch gradient norms using MLflow. Early loss traces (Figure 3) show a steady decrease in the total and CTDN loss terms; however, the content loss remains stagnant and low, indicating weak learning signals reaching the encoder.

This was further confirmed through gradient monitoring across multiple training regimes. In the fixed-weight setting without curriculum (Figure 4), CTDN gradients overwhelmingly dominated, while the encoder and decoder gradients remained close to zero throughout.

After incorporating GradNorm and a curriculum-based training schedule (Figure 5), encoder and decoder gradients revived and became more balanced. However, further experiments with overly aggressive CTDN loss reweighting (Figure 6) caused unstable and noisy encoder gradients, emphasizing the sensitivity of the system to decomposition loss design.

Table 5: Stage 2 loss weight sweep. Performance across values of γ and λ_{scc} .

γ	λ_{scc}	PSNR $\uparrow$	SSIM $\uparrow$	NIQE $\downarrow$	PI $\downarrow$
0.1	0.001	13.03	0.393	73.69	37.30
0.5	0.001	13.03	0.393	69.40	35.16
0.2	0.001	13.03	0.393	76.28	38.60

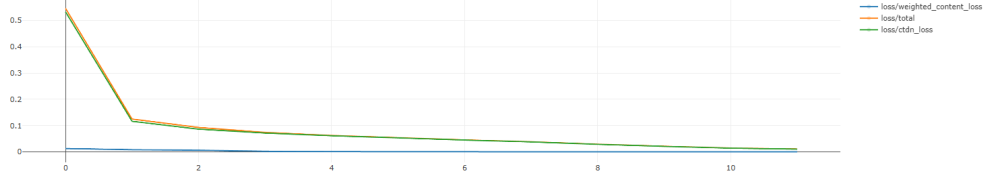


Figure 3: Training loss over time. Despite total loss reduction, the content loss fails to recover in fixed-weight training.

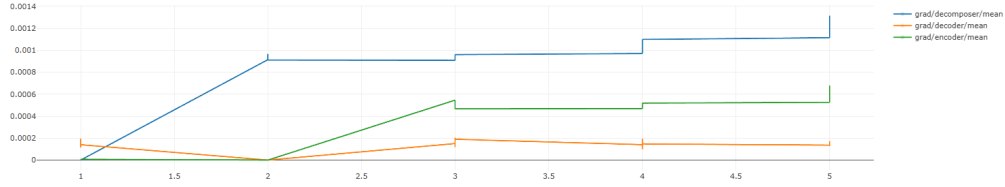


Figure 4: Gradient norms (fixed weights, no curriculum): CTDN dominates, starving other branches.

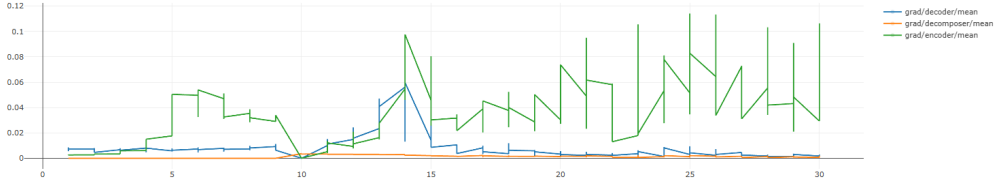


Figure 5: With GradNorm and curriculum: encoder and decoder gradients begin recovering around mid-training.

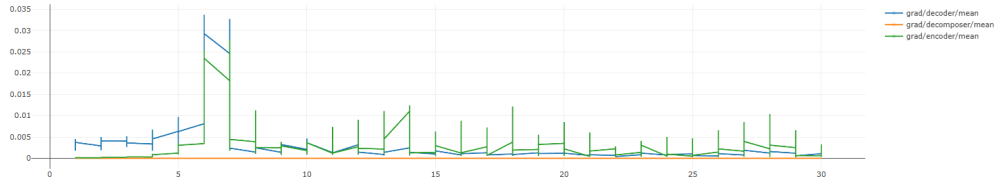


Figure 6: Over-corrected decomposition sweep: encoder gradients spike early but remain unstable.

4.8 Evolving the Training Strategy: Curriculum and GradNorm

These diagnostics directly shaped the development of a three-phase curriculum training strategy:

1. Content-only training (CTDN frozen)
2. CTDN-only training (decoder frozen)
3. Joint fine-tuning with GradNorm-based loss balancing

We further refined this into a flexible, adaptive scheduler with:

- Separate optimizers for encoder, decoder, and decomposer
- Warm-started GradNorm scaling parameters
- Plateau-detection heuristics for transitioning between phases

This strategy improved gradient flow consistency and offered insight into convergence behavior.

4.9 Qualitative Observations: Signs of Encoder Recovery

Although Stage 1 remains a bottleneck, later experiments show evidence of encoder recovery. Early fixed-weight runs produced grayscale and collapsed encodings (Figure 7). Under the curriculum-enhanced GradNorm schedule, encoder activations regained color structure, and reflectance maps began to stabilize (Figure 8).

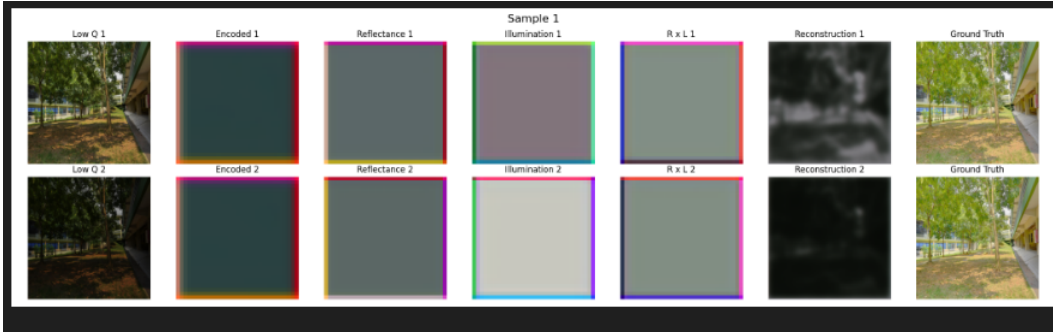


Figure 7: Encoder produce grayscale and collapsed encodings before curriculum learning.

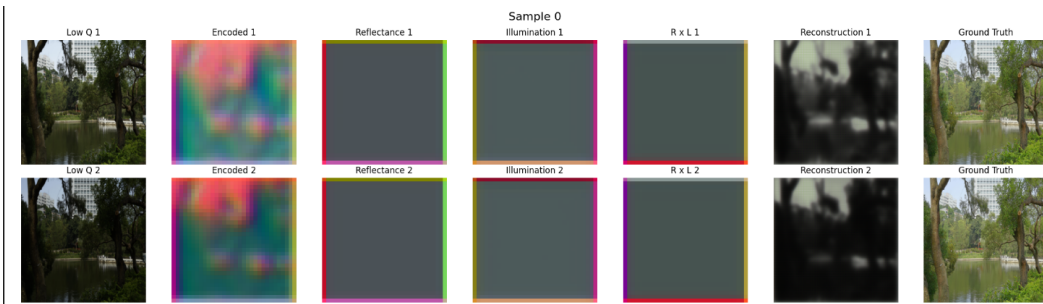


Figure 8: Encoder show structure and color after curriculum learning.

4.10 Conclusion of Experiments

Our experiments highlight several actionable insights into training the LightenDiffusion framework under low-data and constrained-resource conditions:

- **Stage 1 decomposition instability is the core training bottleneck**, with encoder gradients often suppressed by dominating CTDN supervision.
- **Manual loss weighting is insufficient**, showing negligible impact on reconstruction fidelity and high sensitivity to minor changes, especially in decomposition weights.
- **Curriculum learning and GradNorm together offer a viable path forward**, enabling dynamic rebalancing of task importance and staged signal delivery to model components.
- **However, curriculum-based schedules incur training overhead**, requiring careful tuning of phase transitions, warm-up heuristics, and the management of separate optimizers per sub-network.

5 Conclusion

In this report, we developed and rigorously evaluated a two-stage framework for unsupervised low-light image enhancement, inspired by the latent-Retinex diffusion formulation. While the original architecture showed promise on large-scale data, our experiments revealed severe training instability when jointly optimizing for content reconstruction and decomposition on smaller datasets like SICE.

To mitigate this, we introduced a structured three-phase curriculum that progressively activates different loss components—starting with content reconstruction, then decomposition, and finally joint optimization. This curriculum, combined with GradNorm-based dynamic loss weighting, stabilized training and improved generalization without relying on manual hyperparameter sweeps.

Despite these improvements, training remains sensitive to phase scheduling and decomposition weights. Furthermore, encoder and decoder gradients often remained suppressed in early experiments, revealing a bottleneck in Stage 1 learning. Our later iterations showed that decoupling gradient flow across modules—via separate optimizers and adaptive transitions—partially resolves this, but at the cost of longer training time.

Limitations. The system is still vulnerable to instability when exposed to severely under-exposed or texture-less regions. Additionally, the effectiveness of curriculum and GradNorm is contingent on hand-crafted heuristics such as ramp lengths and patience thresholds.

Future Work. Promising directions include: (1) replacing fixed curriculum schedules with automatic phase switching based on plateau detection or gradient dynamics; (2) pretraining the encoder and decoder on auxiliary tasks to bootstrap feature representations and mitigate early gradient starvation; (3) leveraging semantic priors or spatial constraints in Stage 1 to further guide decomposition;

References

- [1] et al. Jianghai. Lightendiffusion: Unsupervised low-light image enhancement with latent-retinex diffusion models. <https://github.com/JianghaiSCU/LightenDiffusion>, 2024. ECCV 2024.
- [2] Edwin H Land. The retinex theory of color vision. *Scientific American*, 237(6):108–129, 1977.
- [3] Xiaojie Guo and Yu Li. Lime: Low-light image enhancement via illumination map estimation. In *IEEE Transactions on Image Processing*, volume 26, pages 982–993, 2016.
- [4] Yonghua Zhang, Jiawan Zhang, and Xiaojie Guo. Kindling the darkness: A practical low-light image enhancer. In *ACM International Conference on Multimedia*, pages 1632–1640, 2019.
- [5] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Deep retinex decomposition for low-light enhancement. In *British Machine Vision Conference (BMVC)*, 2018.
- [6] Yonghua Zhang, Xintong Di, Bin Zhang, and Chunhui Zhang. Beyond brightening low-light images. In *International Journal of Computer Vision*, volume 129, pages 1013–1037, 2021.
- [7] Yifan Jiang, Xinyu Gong, Ding Liu, Yu Cheng, Chen Fang, Xiaohui Shen, Jianchao Yang, Pan Zhou, and Zhangyang Wang. Enlightengan: Deep light enhancement without paired supervision. *IEEE Transactions on Image Processing*, 30:2340–2349, 2021.

- [8] Chunle Guo, Chongyi Li, Jichang Guo, Chen Change Loy, Junhui Hou, and Sam Kwong. Zero-reference deep curve estimation for low-light image enhancement. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1780–1789, 2020.
- [9] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 6840–6851, 2020.
- [10] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations (ICLR)*, 2020.
- [11] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *International Conference on Machine Learning (ICML)*, pages 41–48, 2009.
- [12] Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International Conference on Machine Learning (ICML)*, pages 794–803, 2018.